TECHNICAL SPECIFICATION

ISO/TS 19488

First edition
2021-04

Acoustics — Acoustic classification of dwellings

Acoustique — Système de classification acoustique des logements

Reference number
ISO/TS 19488:2021(E)

© ISO 2021



COPYRIGHT PROTECTED DOCUMENT

© ISO 2021
All rights reserved. Unless otherwise specified, or required in the context of its implementation, no part of this publication may be reproduced or utilized otherwise in any form or by any means, electronic or mechanical, including photocopying, or posting on the internet or an intranet, without prior written permission. Permission can be requested from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office
CP 401 • Ch. de Blandonnet 8
CH-1214 Vernier, Geneva
Phone: +41 22 749 01 11
Email: copyright@iso.org
Website: www.iso.org
Published in Switzerland

Contents

Page

Foreword ..... **iv**

Introduction ..... **v**

1 Scope **1**

2 Normative references **1**

3 Terms and definitions **2**

3.1 General 2

3.2 Classes and single-number quantities to express building performance 4

4 Acoustic classes **6**

5 Classification criteria **7**

5.1 General requirements 7

5.2 Airborne sound insulation 7

5.3 Impact sound insulation 7

5.4 Airborne sound insulation against exterior noise (traffic, industry and other sources) 8

5.5 Sound from building service equipment 9

5.6 Reverberation time and sound absorption in stairwells and access areas 9

6 Verification **9**

Annex A (informative) **Verification of conformity with an acoustic class** **11**

Annex B (informative) **Explanation of intent of classes** **15**

Bibliography **16**

© ISO 2021 – All rights reserved

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see ~~www.iso.org/directives~~).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see ~~www.iso.org/patents~~).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see ~~www.iso .org/iso/foreword .html~~.

This document was prepared by Technical Committee 43, *Acoustics*, Subcommittee SC 2, *Building acoustics*.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at ~~www.iso .org/members .html~~.

© ISO 2021 – All rights reserved

Introduction

Building regulations typically specify requirements about acoustic conditions for new dwellings. However, complying with such requirements does not guarantee satisfactory conditions for the occupants. Thus, there is a need for a guideline with acoustic classes reflecting higher levels of acoustic comfort in new housing. In addition, this document provides a useful tool for characterizing acoustic conditions in older housing and for specifying the goal for acoustic upgrading, when renovating.

The classification guideline specifies criteria for six classes A, B, C, D, E and F for dwellings, class A being the highest class and F the lowest class. If no acoustic performance is required, or the performance is outside the indicated classes or not determined, it can be declared as NPD (no performance determined).

© ISO 2021 – All rights reserved

Acoustics — Acoustic classification of dwellings

1 Scope

This document describes criteria and procedures for acoustic classification of dwellings.

The purpose of this document is to make it easier for developers to specify a classified level of acoustic quality for a dwelling, and help users and builders to be informed about the acoustic conditions and define increased acoustic quality. The document can also be applied as a general tool to characterize the quality of the existing housing stock and includes provisions for classifying the acoustic quality before and after renovation has taken place. By the acoustic quality for a dwelling is understood the quality of the acoustic performances typically included in building regulations, e.g. sound insulation towards neighbouring premises and road traffic as well as sound from service equipment. Sound insulation and room acoustics internally in a dwelling are not included in the acoustic classes defined.

This document does not have a legal status in a country, unless decided by its own authorities. However, an additional purpose of this document is to help national authorities and standardization organisations to develop or revise national building regulations and acoustic classification schemes.

For the purpose of this document, the term "dwellings" refers to detached and attached dwelling-houses, buildings with several flats as well as individual dwellings, and a dwelling is the living space for a household.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 717-1, *Acoustics — Rating of sound insulation in buildings and of building elements — Part 1: Airborne sound insulation*

ISO 717-2, *Acoustics — Rating of sound insulation in buildings and of building elements — Part 2: Impact sound insulation*

ISO 1996-1, *Acoustics — Description, measurement and assessment of environmental noise — Part 1: Basic quantities and assessment procedures*

ISO 1996-2, *Acoustics — Description, measurement and assessment of environmental noise — Part 2: Determination of sound pressure levels*

ISO 3382-2, *Acoustics — Measurement of room acoustic parameters — Part 2: Reverberation time in ordinary rooms*

ISO 10052, *Acoustics — Field measurements of airborne and impact sound insulation and of service equipment sound — Survey method*

ISO 12354-1, *Building acoustics — Estimation of acoustic performance of buildings from the performance of elements — Part 1: Airborne sound insulation between rooms*

ISO 12354-2, *Building acoustics — Estimation of acoustic performance of buildings from the performance of elements — Part 2: Impact sound insulation between rooms*

ISO 12354-3, *Building acoustics — Estimation of acoustic performance of buildings from the performance of elements — Part 3: Airborne sound insulation against outdoor sound*

© ISO 2021 – All rights reserved

EN 12354-5, *Building acoustics — Estimation of acoustic performance of building from the performance of elements — Part 5: Sounds levels due to the service equipment*

EN 12354-6, *Building acoustics — Estimation of acoustic performance of buildings from the performance of elements — Part 6: Sound absorption in enclosed spaces*

ISO 16032, *Acoustics — Measurement of sound pressure level from service equipment in buildings — Engineering method*

ISO 16283-1, *Acoustics — Field measurement of sound insulation in buildings and of building elements — Part 1: Airborne sound insulation*

ISO 16283-2, *Acoustics — Field measurement of sound insulation in buildings and of building elements — Part 2: Impact sound insulation*

ISO 16283-3, *Acoustics — Field measurement of sound insulation in buildings and of building elements — Part 3: Façade sound insulation*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at ~~https://www.iso.org/obp~~

— IEC Electropedia: available at ~~http://www.electropedia.org/~~

3.1 General

3.1.1
airborne sound insulation
insulation against airborne sound measured in accordance with ISO 16283-1 and determined in frequency bands (one-third-octave bands or octave bands) from which single-number ratings for the building performance can be obtained in accordance with ISO 717-1

Note 1 to entry: Single number ratings are expressed in decibels.

Note 2 to entry: Conversion between different single number quantities, e.g. between R'_w and $D_{nT,w}$, is described in ISO 12354-1.

3.1.2
impact sound insulation
insulation against impact sound measured as an impact sound pressure level in accordance with ISO 16283-2 and determined in frequency bands (one-third octave bands or octave bands) from which single-number ratings for the building performance can be obtained in accordance with ISO 717-2

Note 1 to entry: Single number ratings are expressed in decibels.

Note 2 to entry: In ISO 717-2, it is referred to as impact sound pressure level.

Note 3 to entry: Conversion between different single number quantities, e.g. between $L'_{n,w}$ and $L'_{nT,w}$, is described in ISO 12354-2.

3.1.3
airborne sound insulation of building envelopes
insulation against sound from the outside measured in accordance with ISO 16283-3 and determined in frequency bands (one-third-octave bands or octave bands) from which single-number ratings for the building performance can be obtained in accordance with ISO 717-1

Note 1 to entry: Single number ratings are expressed in decibels.

© ISO 2021 – All rights reserved

Note 2 to entry: The application of spectrum adaptation terms C and C_{tr} for different types of sound source is described in ISO 717-1.

Note 3 to entry: The building envelope includes all façade components, including windows, air intakes and roofs.

3.1.4
day-evening-night sound pressure level
L_{den}

weighted sound pressure level as defined in ISO 1996-1 and determined in accordance with ISO 1996-2

$$L_{den} = 10\lg\left\{\frac{1}{24}\left(t_{day}\,10^{0,1\,L_{da}} + t_{evening}\,10^{0,1\,(L_{evenin}+5)} + t_{night}\,10^{0,1\,(L_{night}+10)}\right)\right\}$$

where

L_{den}, L_{day}, $L_{evening}$ and L_{night} are expressed in decibels (dB),

t_{day}, $t_{evening}$ and t_{night} are expressed in hours,

L_{day}, $L_{evening}$ and L_{night} are for the periods t_{day}, $t_{evening}$ and t_{night}, respectively, and

$t_{day} + t_{evening} + t_{night}$ = 24 h.

Note 1 to entry: The default values for t_{day}, $t_{evening}$ and t_{night} are 12 h, 4 h and 8 h, respectively, but individual countries may define these time periods differently.

3.1.5
sound pressure level from service equipment
sound pressure level in a room due to the typical operation of service equipment in a building

Note 1 to entry: It is expressed in decibels.

Note 2 to entry: The measurement result is given as a single-number quantity, A-weighted time-averaged sound pressure level and/or the A-weighted maximum sound pressure level using time-weighting Fast (F) or Slow (S) during a specified working cycle of considered equipment.

3.1.6
service equipment
permanent outdoor and indoor technical installations of a building, such as ventilation systems, lifts, heating systems, cooling systems, emergency power supplies, sanitary installations, central vacuum cleaner, heat pumps, motorized equipment like roller shutters and garage doors, mechanical door closers, internal rainwater pipes, and other similar installations necessary for operation of the building

3.1.7
reverberation time
T

time that is required for the sound pressure level to decrease by 60 dB after the sound source has stopped

Note 1 to entry: T is expressed in seconds.

Note 2 to entry: Measurements are carried out according to ISO 3382-2.

3.1.8
equivalent absorption area of a room
A

hypothetical area of a totally absorbing surface without diffraction effects which, if it were the only absorbing element in the room, would give the same *reverberation time* (~~3.1.7~~) as the room under consideration

Note 1 to entry: A is expressed in m^2.

© ISO 2021 – All rights reserved

3.1.9
walkable surface
S_{floor}
total surface that can be trodden as measured in a plan view and eventually summed over several storeys within the same space

Note 1 to entry: The surface refers to a top view.

Note 2 to entry: S_{floor} is expressed in m^2.

3.1.10
habitable room
room which provides the accommodation of the dwelling, including living room, dining room, study, home office, conservatory, bedroom, etc., excluding bathroom, WC, utility room, store room and circulation space

3.1.11
access area
any type of area which is accessible to all habitants in the building, such as stairwells, corridors, central lobby areas, passageways and other communal areas shared with other units

3.2 Classes and single-number quantities to express building performance

3.2.1
class
category specifying a defined level of acoustic conditions in dwellings

Note 1 to entry: The classes are A, B, C, D, E, F and NPD.

Note 2 to entry: The indication NPD (no performance determined) can be used for dwellings where no acoustic performance is required or determined, or if the performance does not comply with class F.

3.2.2
weighted standardized level difference
$D_{nT,w}$
difference in the space and time average sound pressure levels in two rooms produced by one or more sound sources in one of them, corresponding to a reference value of the reverberation time in the receiving room in accordance with ISO 16283-1 and weighted to a single number according to ISO 717-1

Note 1 to entry: $D_{nT,w}$ is expressed in decibels.

3.2.3
standardized A-weighted level difference
$D_{nT,A}$
sum of the *weighted standardized level difference* (~~3.2.2~~), $D_{nT,w}$, and the spectrum adaptation term, C, according to ISO 717-1 ($D_{nT,A} = D_{nT,w} + C$)

Note 1 to entry: $D_{nT,A}$ is expressed in decibels.

3.2.4
standardized A-weighted level difference extended to low frequencies
$D_{nT,50}$
sum of the *weighted standardized level difference* (~~3.2.2~~), $D_{nT,w}$, and the spectrum adaptation term, $C_{50\text{-}3150}$, according to ISO 717-1 ($D_{nT,50} = D_{nT,w} + C_{50\text{-}3150}$)

Note 1 to entry: $D_{nT,50}$ is expressed in decibels.

© ISO 2021 – All rights reserved

3.2.5
weighted standardized impact sound pressure level

$L'_{nT,w}$

space and time average sound pressure level, when the floor under test is excited by the standardized tapping machine corresponding to a reference value of the reverberation time in the receiving room in accordance with ISO 16283-2 and weighted to a single number according to ISO 717-2

Note 1 to entry: $L'_{nT,w}$ is expressed in decibels.

3.2.6
weighted standardized impact sound pressure level extended to low frequencies

$L'_{nT,50}$

sum of the *weighted standardized impact sound pressure level* (~~3.2.5~~), $L'_{nT,w}$, and the spectrum adaptation term, $C_{I,50\text{-}2500}$, according to ISO 717-2 ($L'_{nT,50} = L'_{nT,w} + C_{I,50\text{-}2500}$)

Note 1 to entry: $L'_{nT,50}$ is expressed in decibels.

3.2.7
weighted standardized level difference for a building envelope

$D_{nT,A,tr}$

sum of the *weighted standardized level difference* (~~3.2.2~~) between the sound pressure level outdoor at 2 m in front of the building envelope and the sound pressure level in the receiving room, corresponding to a reference value of the reverberation time in accordance with ISO 16283-3 and weighted to a single number according to ISO 717-1, and the spectrum adaptation term, C_{tr}, according to ISO 717-1 ($D_{nT,A,tr} = D_{2m,nT,w} + C_{tr}$)

Note 1 to entry: $D_{nT,A,tr}$ is expressed in decibels. It may also be expressed as $D_{2m,nT,A,tr}$.

3.2.8
standardized A-weighted equivalent sound pressure level

$L_{A,eq,nT}$

standardized equivalent A-weighted sound pressure level in a room, due to the sound produced by service equipment in the building, measured according to ISO 10052 or ISO 16032

Note 1 to entry: $L_{A,eq,nT}$ is expressed in decibels.

Note 2 to entry: This sound pressure level is obtained from the equivalent sound pressure level in octave bands, L_{eq}, from 63 Hz to 4 kHz. The sound pressure levels in octave bands are standardized, $L_{eq,nT}$.

Note 3 to entry: Direct reading of an A-weighted sound pressure level and simplified standardization (correction for reverberation time from a table) may be made for survey purposes according to ISO 10052, but in case of dispute, the engineering method in ISO 16032 with measured reverberation times is the reference method.

3.2.9
standardized A-weighted maximum sound pressure level with F time weighting

$L_{AF,max,nT}$

standardized and A-weighted maximum sound pressure level using time weighting "F" in a room, due to the sound produced by *service equipment* (~~3.1.6~~) in the building, measured according to ISO 10052 or ISO 16032

Note 1 to entry: $L_{AF,max,nT}$ is expressed in decibels.

Note 2 to entry: When this sound pressure level is determined according to ISO 10052, it is obtained by direct reading of an A-weighted maximum sound pressure level or from the maximum sound pressure level in octave bands from 63 Hz to 4 kHz using time weighting "F". The sound pressure levels are standardized, $L_{F,max,nT}$.

© ISO 2021 – All rights reserved

3.2.10
standardized A-weighted maximum sound pressure level with S time weighting

$L_{AS,max,nT}$
standardized and A-weighted maximum sound pressure level using time weighting "S" in a room, due to the sound produced by *service equipment* (~~3.1.6~~) in the building, measured according to ISO 10052 or ISO 16032

Note 1 to entry: $L_{AS,max,nT}$ is expressed in decibels.

Note 2 to entry: When this sound pressure level is determined according to ISO 10052, it is obtained by direct reading of an A-weighted maximum sound pressure level or from the maximum sound pressure level in octave bands from 63 Hz to 4 kHz using time weighting "S". The sound pressure levels are standardized, $L_{S,max,nT}$.

4 Acoustic classes

The classes A to F specify different levels of acoustic conditions in dwellings. Class A is the highest class, class F is the lowest class. The intended meaning of classes is explained in ~~Annex B~~. The indication NPD can be used for dwellings where no acoustic performance is required or determined, or if the performance does not meet the requirements of class F.

The classification includes criteria for the six classes A-F for each of the following five acoustic aspects:

— airborne sound insulation;

— impact sound insulation;

— airborne sound insulation of building envelopes against outdoor noise from traffic, industry or other sources;

— sound pressure levels in the dwellings from service equipment;

— reverberation time or ratio of the equivalent sound absorption area to the walkable surface in common access areas or stairwells and corridors with dwellings opening onto them.

The airborne sound insulation between rooms of a building characterizes the ability to insulate against airborne sound transmission between the rooms.

The impact sound insulation characterizes the extent to which a building construction transfers impact sound within the building.

The airborne sound insulation of building envelope characterizes the building envelope's ability to insulate against airborne sound transmission into a building.

The sound pressure level from service equipment characterizes the received sound pressure level in a room due to the operation of a specific piece of service equipment applied for operation of the building.

The reverberation time and sound absorption describe the room acoustic characteristics of access areas and common stairwells.

Steps between classes for sound insulation and noise levels are generally 4 dB (unless the single number quantity changes), and the total range is thus approximately 20 dB. Furthermore, class limits for reverberation time or sound absorption for stairwells and common access areas are included as a part of classification of dwellings or buildings.

All requirements for a class shall be fulfilled in order to obtain a certain class designation. A classification can be made for a whole residential building only if all dwellings in the building fulfil the class criteria. Dwellings in a building can also be assigned different classes. Classification can also be made for an individual dwelling or a specific characteristic, e.g. airborne sound insulation, separately. The classification applies as long as there are no adverse changes in building constructions or environment. If such changes have occurred, e.g. by new roads nearby, the classification should be reconsidered. Sound insulation between rooms internally in a dwelling is not included in the classification.

© ISO 2021 – All rights reserved

The classification of a dwelling or a residential building is in principle based on sample measurements in accordance with the verification procedure defined in ~~Clause 6~~ of this document. In the design stage, an estimate can be made on the basis of predictions and experiences from similar building constructions. Such predicted performance values combined with site inspections can be included in the verification procedure.

5 Classification criteria

5.1 General requirements

Dwellings to be classified according to this document shall provide appropriate protection against sound from other parts of the same building, from adjoining buildings and from the outdoor environment, according to the intent of each acoustic class.

Descriptions and definitions of parameters applied in this document are found in ~~Clause 3~~.

5.2 Airborne sound insulation

In ~~Table 1~~, the minimum values for airborne sound insulation between dwellings and other dwellings as well as between premises with noisy activities and dwellings are shown for each of the classes.

Table 1 — Airborne sound insulation between dwellings and other rooms — Class limits[a]

Type of space		Class A dB	Class B dB	Class C dB	Class D dB	Class E dB	Class F dB
1	Between habitable rooms in a dwelling and rooms outside the dwelling in all directions	$D_{nT,50} \geq 58$	$D_{nT,50} \geq 54$	$D_{nT,A} \geq 52$	$D_{nT,A} \geq 48$	$D_{nT,A} \geq 44$	$D_{nT,A} \geq 40$
2	From common stairwells or access areas into habitable rooms in dwellings, where there is an entrance door in the separating wall	$D_{nT,A} \geq 46$	$D_{nT,A} \geq 42$	$D_{nT,A} \geq 38$	$D_{nT,A} \geq 34$	$D_{nT,A} \geq 30$	$D_{nT,A} \geq 26$
3	From premises with noisy activities into habitable rooms in dwellings [b]	$D_{nT,50} \geq 64$	$D_{nT,50} \geq 60$	$D_{nT,A} \geq 58$	$D_{nT,A} \geq 54$	$D_{nT,A} \geq 50$	$D_{nT,A} \geq 46$

[a] Different descriptors are applied to reflect use of different frequency ranges and weightings. Instead of $D_{nT,A}$, $D_{nT,w}$ may be applied if 2 dB is added to the limit value. If $D_{nT,A}$ is applied instead of $D_{nT,50}$, 2 dB should be added to the limit value of $D_{nT,50}$. For comparison between descriptors, see ISO 12354-1.

[b] Premises with noisy activities are rooms for shared services like laundries, central boiler house, joint/commercial kitchens or commercial premises like shops, workshops or cafés. However, in each case, noise levels should be estimated and the sound insulation designed accordingly, e.g. for party rooms, discotheques, etc. Then, the limits given in ~~Table 4~~ for service equipment sound can be used as design goals.

5.3 Impact sound insulation

In ~~Table 2~~, the maximum values for impact sound pressure level in dwellings from other dwellings, from premises with noisy activities, and from common stairwells and access areas as well as balconies, terraces, bath, toilet not belonging to own dwellings are shown for each of the classes.

© ISO 2021 – All rights reserved

Table 2 — Impact sound pressure level in dwellings — Class limits

Type of space		Class A dB	Class B dB	Class C dB	Class D dB	Class E dB	Class F dB
1	In habitable rooms in dwell- ings from other dwellings in all directions	$L'_{nT,w} \leq 46$ and $L'_{nT,50} \leq 50$ [a]	$L'_{nT,w} \leq 50$ and $L'_{nT,50} \leq 54$ [a]	$L'_{nT,w} \leq 54$	$L'_{nT,w} \leq 58$	$L'_{nT,w} \leq 62$	$L'_{nT,w} \leq 66$
2	In habitable rooms in dwellings from: — common stairwells or access areas — balconies or terraces or bath rooms not belonging to own dwelling [b]	$L'_{nT,w} \leq 50$	$L'_{nT,w} \leq 54$	$L'_{nT,w} \leq 58$	$L'_{nT,w} \leq 62$	$L'_{nT,w} \leq 66$	$L'_{nT,w} \leq 70$
3	In habitable rooms in dwell- ings from premises with noisy activities [c]	$L'_{nT,w} \leq 40$ and $L'_{nT,50} \leq 44$ [a]	$L'_{nT,w} \leq 44$ and $L'_{nT,50} \leq 48$ [a]	$L'_{nT,w} \leq 48$	$L'_{nT,w} \leq 52$	$L'_{nT,w} \leq 56$	$L'_{nT,w} \leq 60$

[a] Experience has shown that when applying the low-frequency rating, potentially disturbing high frequency sounds are not rated appropriately, and for this reason, two descriptors are applied in order to account for both hard floor impact sounds as well as low frequency footstep sounds. The limit values for $L'_{nT,w}$ are 4 dB lower than those specified for $L'_{nT,50}$. For comparison between descriptors, see ISO 12354-2.

[b] Impact sound from small balconies and rooms (area less than 4 m^2) are not included, e.g. toilets and utility rooms.

[c] Premises with noisy activities are rooms for shared services like laundries, central boiler house, joint/commercial kitchens or commercial premises like shops, workshops or cafés. However, in each case, noise levels shall be estimated and the sound insulation designed accordingly, e.g. for party rooms, discotheques, etc.

5.4 Airborne sound insulation against exterior noise (traffic, industry and other sources)

The sound insulation aims at providing acoustic quality in habitable rooms. Sound levels indoors depend highly on the outdoor source and sound insulation.

The minimum values of the class for sound insulation are shown in ~~Table 3~~ for a specific environment as characterized by L_{den} for the relevant outdoor sound sources. In case that a dominant source of noise on the outside has a different spectrum than C_{tr}, a more relevant spectrum adaptation term than C_{tr} shall be applied to calculate the weighted single number, $D_{nT,A,tr}$, in order to get about the same sound pressure level in the habitable rooms. The index "tr" may then be replaced by a more descriptive index for the actual type of source.

Table 3 — Sound insulation in dwellings against exterior noise — Class limits [ab]

Type of space		Class A dB	Class B dB	Class C dB	Class D dB	Class E dB	Class F dB
1	Façades and roofs of habit- able rooms in dwellings; in specific environment with sound sources character- ized by L_{den}	$D_{nT,A,tr} \geq L_{den} - 20$	$D_{nT,A,tr} \geq L_{den} - 24$	$D_{nT,A,tr} \geq L_{den} - 28$	$D_{nT,A,tr} \geq L_{den} - 32$	$D_{nT,A,tr} \geq L_{den} - 36$	$D_{nT,A,tr} \geq L_{den} - 40$

[a] The sound insulation values are expressed as a weighted standardized sound level difference with a spectrum adaptation term for road traffic noise. For other types of sound source than road traffic noise, $D_{nT,A,tr}$ shall be determined from the relevant level and spectrum of the sources. $D_{nT,w} + C_{tr,50-3150}$ may be used where low frequency sound influences the indoor sound pressure level. e.g. where the sound comes from mechanical equipment placed outside the building.

[b] $D_{nT,A,tr} \geq 30$ dB applies as a minimum requirement to classes A–D.

© ISO 2021 – All rights reserved

5.5 Sound from building service equipment

The maximum values of the classes for A-weighted time-averaged or the maximum sound pressure levels due to service equipment are shown in Table 4.

Table 4 — Sound pressure levels in dwellings due to building service equipment — Class limits

Type of space and sources [a]		Quantity	Class A dB	Class B dB	Class C dB	Class D dB	Class E dB	Class F dB
1	In habitable rooms in dwellings from outdoor and indoor service equipment producing continuous sound	$L_{A,eq,nT}$	≤22	≤26	≤30	≤34	≤38	≤42
2	In habitable rooms in dwellings from outdoor and indoor service equipment producing intermittent or irregular sound, from neighbouring spaces	$L_{AF,max,nT}$ [c]	≤26	≤30	≤34	≤38	≤42	≤46

[a] Requirements relate to sounds that occur more than occasionally due to service equipment in neighbouring dwellings, equipment serving the whole building and service equipment within the dwelling for normal ventilation/heating/cooling.

[b] Sound with tonal components can be perceived more annoying and be subject to national regulations.

[c] $L_{AS,max,nT}$ may also be used, provided that 4 dB stricter limits (lower sound levels) are fulfilled, i.e. the same as $L_{Aeq,nT}$.

5.6 Reverberation time and sound absorption in stairwells and access areas

The maximum values for reverberation time and minimum values of equivalent sound absorption area in stairwells and access areas adjacent to habitable rooms are shown in Table 5.

Table 5 — Reverberation time *T* and sound absorption *A* — Class limits

Type of space		Class A	Class B	Class C	Class D	Class E	Class F
1	In access areas (except common stairwells) [a]	$T \leq 0{,}6$ s	$T \leq 0{,}9$ s	$T \leq 1{,}2$ s	$T \leq 1{,}5$ s	$T \leq 1{,}8$ s	$T \leq 2{,}1$ s
2	In common stairwells [ab]	$T \leq 0{,}9$ s or $A \geq 0{,}45 \times S_{floor}$	$T \leq 1{,}2$ s or $A \geq 0{,}35 \times S_{floor}$	$T \leq 1{,}5$ s or $A \geq 0{,}25 \times S_{floor}$	$T \leq 1{,}8$ s or $A \geq 0{,}20 \times S_{floor}$	$T \leq 2{,}1$ s or $A \geq 0{,}15 \times S_{floor}$	$T \leq 2{,}4$ s or $A \geq 0{,}10 \times S_{floor}$

[a] The limits for reverberation time are maximum values, and the limits for equivalent sound absorption are minimum values, in both cases for each of the octave bands 500 Hz, 1000 Hz and 2000 Hz.

[b] For practical reasons, as an alternative to the requirement of a reverberation time in this type of spaces, a corresponding amount of equivalent sound absorption area according to EN 12354–6 has been added, using the equivalent absorption area $A \geq 0{,}16\ V/T$ and an approximately 0,3 s longer *T*-value, compared to the first row.

6 Verification

Verification shall be based on a number of measurements to characterize each aspect of the acoustic classes in the completed building or unit. The rooms for measurements should be selected in accordance with Annex A. The measurements shall be made according to the following standardized methods as specified in Clause 2.

Measure the airborne sound insulation between rooms according to ISO 16283-1 and determine the single numbers quantities in Table 1 from the measurement results according to ISO 717-1.

Measure the impact sound insulation according to ISO 16283-2 and determine the single numbers quantities in Table 2 from the measurement results according to ISO 717-2.

© ISO 2021 – All rights reserved

Determine the day-evening-night sound pressure level, L_{den}, according to ISO 1996-1 and ISO 1996-2. Measure the sound insulation against exterior noise according to ISO 16283-3 and determine the single number quantities in Table 3 from the measurement results according to ISO 717-1.

Measure the sound from building service equipment according to either the survey method in ISO 10052 or the engineering method in ISO 16032 (which is the reference method in case of dispute) and compare with the criteria in Table 4.

Measure the reverberation time in Table 5 according to ISO 3382-2 or estimate the equivalent sound absorption area in Table 5 from the declared sound absorption of the products installed in the building or the unit.

As explained in Annex A, the number of measurements may be reduced, when the verification includes comparisons to similar constructions or theoretical calculations or as a combination thereof, provided visual inspections are carried out during the construction process to confirm the design is implemented as planned. Calculations shall be made according to ISO 12354-1, ISO 12354-2, ISO 12354-3, EN 12354-5 and EN 12354-6.

Theoretical evaluations, experiences with similar building designs as well as visual inspections of critical constructions in the building during production have proved to be valuable as a part of a verification procedure. In cases with several types of constructions, considerable variations in floor plans or use of rooms, it can be necessary to increase the number of measurements. Hence, the number of sample measurements in the completed building should be chosen to fit to the actual conditions. Annex A may be followed in detail or adapted somewhat if needed to reflect special conditions in the building or for other relevant reasons, as long as such changes are not in conflict with the general requirement. It is advised to agree on a complete verification procedure at an early stage of a building project.

© ISO 2021 – All rights reserved

Annex A
(informative)

Verification of conformity with an acoustic class

A.1 General

The aim of this annex is to facilitate verification of conformity with an acoustic class. It should be applied to habitable rooms, i.e. rooms for social activities, rest or sleep, e.g. living rooms and bedrooms. Rooms not being used more than occasionally, e.g. separate kitchens, bathrooms, rooms for storage and similar, may be designed according to the specifications of this document but they are normally not selected for measurements.

Verification can apply to an entire residential building, an individual dwelling or even a specific acoustic characteristic, which are referred to as a "unit" in this annex.

The following conditions should be considered when a unit is subject to an evaluation of conformity with the criteria of a specific acoustic class.

Two alternative verification procedures are described in ~~A.2.1~~ (Procedure A) and ~~A.2.2~~ (Procedure B). Either procedure A or B may be applied. When verifying the acoustic class of a unit, the general principle is that a sufficient number of rooms should be selected in order for the result to represent the unit and each relevant acoustic characteristic should be evaluated for this unit.

Acoustic calculations and measurements are performed according to the relevant standards specified in the main body of this document. Unoccupied (unfurnished) rooms offer favourable measurement conditions, but the results are corrected to represent furnished rooms (as occupied). The persons or organizations that are appointed to make the relevant calculations, visual inspections or measurements shall be qualified for the tasks. The contents of a report of acoustic classification are given in ~~A.3~~.

The classification of a unit applies from a certain date. The classification is valid as long as the building constructions and the outdoor noise conditions remain unaltered. If changes occur, the classification shall be reconsidered.

A.2 Verification of conformity with criteria for an acoustic class

A.2.1 Procedure A: Verification by calculations, visual inspections and field measurements

Verification is made at three stages of development of a new building or a change of an existing building (e.g. retrofit, change of use). The following steps 1 to 3 are required for a complete verification following procedure A:

— Step 1: in the design stage of a building.

  The performance of the proposed constructions can be estimated by experience with similar constructions, or theoretical calculations according to ISO 12354-1, ISO 12354-2 and ISO 12354-3 or EN 12354-5 and EN 12354-6, or as a combination thereof. It is advisable to keep a safety margin of at least half a class (2 dB) for constructions where firm practical experience confirms that this is satisfactory, but larger margins may be applied for other constructions, in particular if these are judged sensitive to workmanship errors, etc. Rooms or constructions which can be sensitive for the classification are to be identified, e.g. with respect to the quality of workmanship or correct operation of a service equipment. Appropriate conclusions should be communicated to the project management and followed up by approval of specifications and construction drawings.

© ISO 2021 – All rights reserved

— Step 2: in the construction stage of a building.

  Visual inspections on site are carried out to clarify whether building elements or service equipment are installed according to the approved specifications and construction drawings.

— Step 3: in the completed building.

  The person or organization appointed for measurements selects 5 % of the structures and rooms to be measured, either randomly or as identified during the design stage. Both a large room and a small room are selected in the normal case.

To achieve the class set as a goal, all measurement results shall in principle meet the criteria of the class in question. However, conformity is granted if the average results comply with class limits and no individual result deviates adversely by more than 2 dB. In very small rooms (with a volume less than 25 m^3), another 2 dB adverse deviations in the airborne and impact sound insulation are acceptable if the single number value is evaluated in the frequency range starting at 50 Hz, because of the low-frequency measurement procedure in ISO 16283 applicable to such rooms.

A reverberation time shall not exceed the requirement by more than 0,15 s, provided the average results over the selected spaces fulfil the requirement.

For service equipment sound, the A-weighted levels may be read-off on site and standardized to 0,5 s reverberation time by a single correction value according to the survey method of ISO 10052. In case of dispute, the more accurate measurements in octave bands according to ISO 16032 are preferred. If background sounds make the sound levels difficult to determine accurately, indicate this in the report with the sign "≤" in front of the result.

A.2.2 Procedure B: Verification by field measurements only

In the completed building, the person or organization appointed for measurements selects at least 10 % of the types of construction of separating walls and floors to be measured in such a way that they are sufficiently representative of the unit. Both a large room and a small room are selected in each dwelling in the normal case.

To facilitate the selection, a quick scan of the entire building or a part thereof can ensure that critical sites/rooms are selected for measurements. For example, partitions with critical flanking constructions or unfavourable locations in the building may be selected, e.g. rooms next to a mechanical room, busy road, etc. More measurements may be added if performance is likely to vary between units.

To achieve the class set as a goal, all measurement results shall in principle meet the criteria of the class in question. However, conformity is granted if the average results comply with class limits and no individual result deviates adversely by more than 2 dB. In very small rooms (with a volume less than 25 m^3), another 2 dB adverse deviations in the airborne and impact sound insulation are acceptable if the single number value is determined by the performance in the frequency range between 50 Hz and 80 Hz, because of the low-frequency measurement procedure in ISO 16283 applicable to such rooms.

A reverberation time may not exceed the requirement by more than 0,15 s, provided the average results for the measurements over the selected spaces fulfil the requirement.

For service equipment sound, the A-weighted levels may be read-off on site and standardized to 0,5 s reverberation time by a single correction value according to the survey method of ISO 10052. In case of dispute, the more accurate measurements in octave bands according to ISO 16032 are preferred. In case background noise makes the sound levels difficult to determine accurately, indicate this in the report with the sign "≤" in front of the result.

A.2.3 Verification of an entire building

When an entire building is to be tested, the number of samples for measurement of airborne and impact-sound insulation and noise levels is described in ~~A.2.1~~ and ~~A.2.2~~. However, the minimum number of measurements for each structure type and acoustic characteristic is always 3.

© ISO 2021 – All rights reserved

A.2.4 Verification of individual dwellings or acoustic characteristic

Individual dwellings or an acoustic characteristic in a building may be classified, if each of them complies with the relevant class limits. The above-mentioned general principles apply.

A.3 Contents of verification report for classification of a dwelling or a building

Reports in which the acoustic classification of a dwelling or a building is presented according to this document shall be uniform and concise. The first page of the report may only contain the most essential information, such as

— the objective of the measurements,

— the client,

— the name of the person(s) or organization responsible for the verification,

— the number of dwellings verified,

— the measurements dates,

— the main results,

— the class obtained,

— other relevant information,

— a reference to this document, i.e. ISO/TS 19488:2021, and

— signatures.

Detailed information about measurement methods, the dwelling(s) or building and the measurement results may be presented in appendices of the classification report. Reference should be made to the specific test reports (with frequency dependent results) being the basis of the class assignment.

If classifications for different dwellings, rooms or acoustic characteristics vary, the classification assigned is the minimum class obtained, when considering all relevant acoustic characteristics. However, additional assignments of higher classes for individual dwellings, rooms or acoustic characteristics can be made according to the rules for such units. In case of different classes for different dwellings, rooms or acoustic characteristics, the report can include overview tables with classes obtained for the different acoustic characteristics and/or for different dwellings or rooms.

At least one of the parameters used in the tables in ~~Clause 5~~ shall be reported, but more parameters may be reported and declared where reference to this document is made. All other requirements of this annex apply for assigning these classifications.

If no acoustic performance is required or the performance is outside the indicated classes or not determined, it can be declared as NPD (no performance determined).

A.4 Classification examples

Four examples are shown below for the presentation of classification results determined according to the procedure in ~~A.2~~. In all cases, the verification report shall include information as described in ~~A.3~~. The four examples are:

a) classification of a residential building with one class for the entire building;

b) classification of a dwelling with one class for the entire dwelling;

c) classification of a residential building with individual class indication for each acoustic characteristic;

© ISO 2021 – All rights reserved

d) classification of a dwelling with individual class indication for each acoustic characteristic.

EXAMPLE 1 A residential building has obtained Class D; since the minimum of this class has been fulfilled for all individual acoustic characteristics.

EXAMPLE 2 A dwelling No. NNN has obtained Class C; since the minimum of this class has been fulfilled for all individual acoustic characteristics for the dwelling.

EXAMPLE 3 An example of classification of a residential building with individual class indication for each acoustic characteristic is given in ~~Table A.1~~.

Table A.1 — Acoustic classification of a residential building

Acoustic characteristic	Class						
	A	B	C	D	E	F	NPD
Airborne sound insulation			X				
Impact sound pressure level				X			
Sound insulation against exterior noise			X				
Sound from building service equipment				X			
Reverberation time (or relative absorption area) in access areas, etc.				X			
The classification result for the entire building is Class D, which is the lowest class for individual acoustic characteristics.							

EXAMPLE 4 An example of classification of a dwelling with individual class indication for each acoustic characteristic is given in ~~Table A.2~~.

Table A.2 — Acoustic classification of dwelling No. NNN in residential building

Acoustic characteristic	Class						
	A	B	C	D	E	F	NPD
Airborne sound insulation			X				
Impact sound pressure level			X				
Sound insulation against exterior noise		X					
Sound from building service equipment			X				
Reverberation time (or relative absorption area) in access areas etc.		X					
The classification result for the entire dwelling is Class C, which is the lowest class for individual acoustic characteristics.							

© ISO 2021 – All rights reserved

Annex B
(informative)

Explanation of intent of classes

The descriptions in ~~Table B.1~~ are based on a pragmatic combination of existing standards and experiences collected from many countries (see Bibliography).

Table B.1 — Description in general terms of the relative acoustic quality of the different classes

Class	General
A	A quiet atmosphere with a high level of protection against intruding sound. This class may be applied where a considerably better acoustic climate than Class C is asked for.
B	Under normal circumstances, a good protection against sound without significant restriction on the behaviour of the neighbours. This class may be applied where a better acoustic climate than Class C is asked for.
C	Protection against significant disturbance, given normal behaviour of neighbours who are considerate of other occupants. Newer building constructions in many countries are likely to fulfil or exceed this class.
D	Disturbance by intruding noise can be expected more than occasionally, even in case of considerate behaviour of neighbours, adjusted to these conditions. Newer building constructions in most countries are likely to fulfil or exceed this class.
E	A low protection is offered against intruding sounds. To be applied mainly for classification of existing housing (before renovation).
F	A very low protection is offered against intruding sounds. To be applied only for classification of older, existing housing (before renovation).
NPD	No performance determined.

NOTE Whether specific sounds are audible depends not only on the building construction but also on the background noise in the room. For higher acoustic classes, less sound is heard and for lower classes, the sound is louder. Even for a high acoustic class, occasionally, loud sounds can still be heard, e.g. from children running and jumping on the floor above or very loud music being played in another dwelling. Whether an audible sound also implies that an occupant is disturbed or annoyed by this sound depends on many factors, e.g. personal habits and activity, feelings about the sound and at which time of the day or night this sound is heard.

© ISO 2021 – All rights reserved

Bibliography

[1] Rindel J.H., Acoustic quality and sound insulation between dwellings. Journal of Building Acoustics. 1999, **5** pp. 291–301

[2] Gerretsen E., Model-based assessment scheme for acoustic quality in buildings. NAG-DAGA, Rotterdam, 2009

[3] Rasmussen B., Machimbarrena M. 2019). *Developing an international acoustic classification scheme for dwellings – From chaos & challenges to compromises & consensus?* Proceedings of InterNoise2019, Madrid, Spain. International Institute of Noise Control Engineering (I-INCE), Noise-Con Proceedings.

[4] Rasmussen B. 2019) *Sound insulation between dwellings – Comparison of national requirements in Europe and interaction with acoustic classification schemes.* Proceedings of ICA 2019, 23rd International Congress on Acoustics, Sept. 2019, Aachen, Germany. Deutsche Gesellschaft für Akustik (DEGA e.V.). ~~https://doi.org/10.18154/RWTH-CONV-239983~~

[5] Rasmussen B. *Acoustic classification of buildings in Europe – Main characteristics of national schemes for housing, schools, hospitals and office buildings*. European Acoustics Association, Proceedings of EuroNoise 2018. Hersonissos, Crete, Greece, 2018

[6] Rasmussen B., Rindel J.H., Sound insulation between dwellings – Descriptors applied in building regulations in Europe. Appl. Acoust. 2010, **71** (3) pp. 171–180. ~~https://doi.org/10.1016/j.apacoust.2009.05.002~~

[7] Rasmussen B., Sound insulation between dwellings – Requirements in building regulations in Europe. Appl. Acoust. 2010, **71** (4) pp. 373–385. ~~https://doi.org/10.1016/j.apacoust.2009.08.011~~

[8] Rindel J.H., Løvstad A., Klæboe R. 2016). Aiming at satisfactory sound conditions in dwellings – the use of dose response curves. BNAM 2016, Stockholm, Sweden. *Proceedings of the Baltic-Nordic Acoustics Meeting 2016, Nordic Acoustics Association.*

© ISO 2021 – All rights reserved

ICS 91.120.20

Price based on 16 pages

© ISO 2021 – All rights reserved